

Application of Imagery Modeling and Deep Learning in Chip Thermal Analysis

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Abstract—Currently, with the reduction of transistor size and the increase in the number of transistors on the chip, thermal simulation in the chip design process has become increasingly important. Thermal simulation can predict the working conditions and temperature of chips in the early stages, providing valuable feedback for chip design to ensure the optimal working temperature. Currently, thermal simulation software typically faces challenges of higher accuracy and longer computation time. The use of deep learning methods helps to quickly obtain high-precision thermal simulation results, which helps to shorten chip design time and accelerate the chip design process. The purpose of the Deep Learning based chip thermal analysis method proposed in this article is not only to verify the feasibility of using deep learning in chip thermal analysis, but also to reduce the time used for thermal simulation in the chip design process and ultimately accelerate the entire chip design process. The experimental dataset is the temperature of multiple layouts of a chip, and using mature image processing deep learning models, high-precision predictions can be achieved.

Keywords—Deep learning, thermal analysis, temperature prediction, imagery of chip layout, FPGA

I. INTRODUCTION

The rapid increase in the number of transistors in chips has led to various serious heating problems during chip operation. The performance of a chip decreases by 4% for every 10 degrees Celsius increase in operating temperature. In addition, the rise in temperature of the chip itself can also increase the risk of chip failure. Therefore, in the semiconductor industry, thermal analysis of chips plays a very important role in every step of chip design. In this process, it is necessary to consider the effects of excessively high peak temperatures and severe thermal gradients on transistor performance, stress, aging, electromigration (EM), voltage drop, and timing [1]. In addition to the necessary thermal simulation guarantee for chip design itself, the arrival of the post Moore's era also means that improving chip performance places more emphasis on better chip layout and wiring design. For example, the maximum temperature difference between different layouts of the same chip during normal operation can reach 10 to 20 °C. This also requires a fast and accurate thermal simulation method to accelerate the working speed.

In the past chip design process, finite element methods were often necessary for chip level thermal analysis, such as software such as Ansys [3] used for chip level thermal simulation. Although the accuracy of finite element method for thermal

simulation is high, the time spent on chip level thermal simulation is too long, which directly leads to an increase in repeated iterations in the chip design process, undoubtedly increasing the working time of chip design. In addition to the finite element method, some researchers have also proposed other methods, such as Hotspot [4] for rapid thermal evaluation, the use of Power Blurring (PB) algorithm [5] for temperature estimation, and the use of Physical Information Neural Network (PINN) [6] to solve heat transfer problems. Including in the field of machine learning [7], there are also applications for researching chip thermal simulation.

In addition to the inherent requirements of integrated circuit design, the development of deep learning technology also provides certain technical support for the research in this article. Deep learning has been a research hotspot in the field of artificial intelligence in recent years, and deep learning technology has achieved excellent results, especially in the field of image recognition. By using deep learning techniques, high-quality analysis of certain image content can be performed, for example. By utilizing deep learning, artificial intelligence drawings can even be completed, such as the currently popular stable diffusion model. This intelligent analysis and design capability is exactly what is needed in the field of integrated circuits.

The main contributions of this article are as follows:

1. A batch of new data based on image-based FPGA chip structures was established, and the feasibility of combining them with deep learning technology for chip thermal analysis was verified.
2. Provide a new fast validation method for thermodynamic optimization of chip layout. The use of deep learning models for fast thermal analysis of chips provides more rapid feedback speed, and generating high-precision thermal simulation data hotspots takes more than 5 minutes, while using image processing methods takes less than 10 seconds.
3. Provides new directions for the application of deep learning in the semiconductor field. By using new modeling approaches for data in the semiconductor field, the application of deep learning has been expanded.

II. METHOD

A. Raw data processing

In the field of deep learning, obtaining qualified datasets is a crucial step, and without appropriate data-driven approaches, successful deep learning models cannot be trained. It is difficult to obtain the required public dataset in the semiconductor field, especially the layout of the required chips and corresponding operating temperature data. The main purpose of this article is to learn the required features based on the layout of the chip itself. Therefore, FPGA chips with relatively distinct layout features are more conducive to image-based models. The layout data of FPGA chips and the corresponding batch acquisition of thermal data can also be quickly obtained with the assistance of many open-source tools, ultimately producing the new dataset required for the entire experiment.

The main tasks of obtaining experimental data in this experiment are as follows:

1. Obtain and search for suitable open-source chip data. After careful consideration, we chose the 2016 chip data from the ISPD competition [9] for the experiment. ISPD2016 is the FPGA chip data of Xilinx, and necessary data such as component power and chip size can be easily obtained through Xilinx technical manual [10] and XPE power estimator.
2. After obtaining the raw data, a method is needed to generate different layouts of the same chip. Considering time constraints, I chose to use the open-source layout and routing software OpenPARF [11] to obtain different layouts for the same chip. OpenPARF is an efficient FPGA location and routing framework designed by Professor Lin Yibo from Peking University. It utilizes multiple electrostatic field methods to achieve optimal results in terms of wiring length and layout speed.
3. After obtaining different layouts of the chip, further thermal simulation is needed to evaluate the heat generation in different layouts. This step needs to be completed before obtaining the required experimental data. Here, The characteristics of Hotspot 6.0 make it easy for me to quickly predict chip layout temperature in batches. Hotspot 6.0 mainly relies on the operating frequency for thermal simulation. By modeling and analyzing the power consumption of the circuit at different frequencies, it can predict the heat distribution in the circuit, which provides an important reference for thermal management and optimization.

B. Framework

At present, the development of deep learning in the field of artificial intelligence is very rapid. Although deep learning has great advantages such as strong feature learning ability and fast running speed, there are significant difficulties in applying it to chip design and feature prediction. If chips are imaged and modeled, various mature image processing models in deep learning can be directly used for chip feature extraction, The

overall experimental framework of this article is based on this idea. The overall framework of this experiment is shown in Fig. 1:

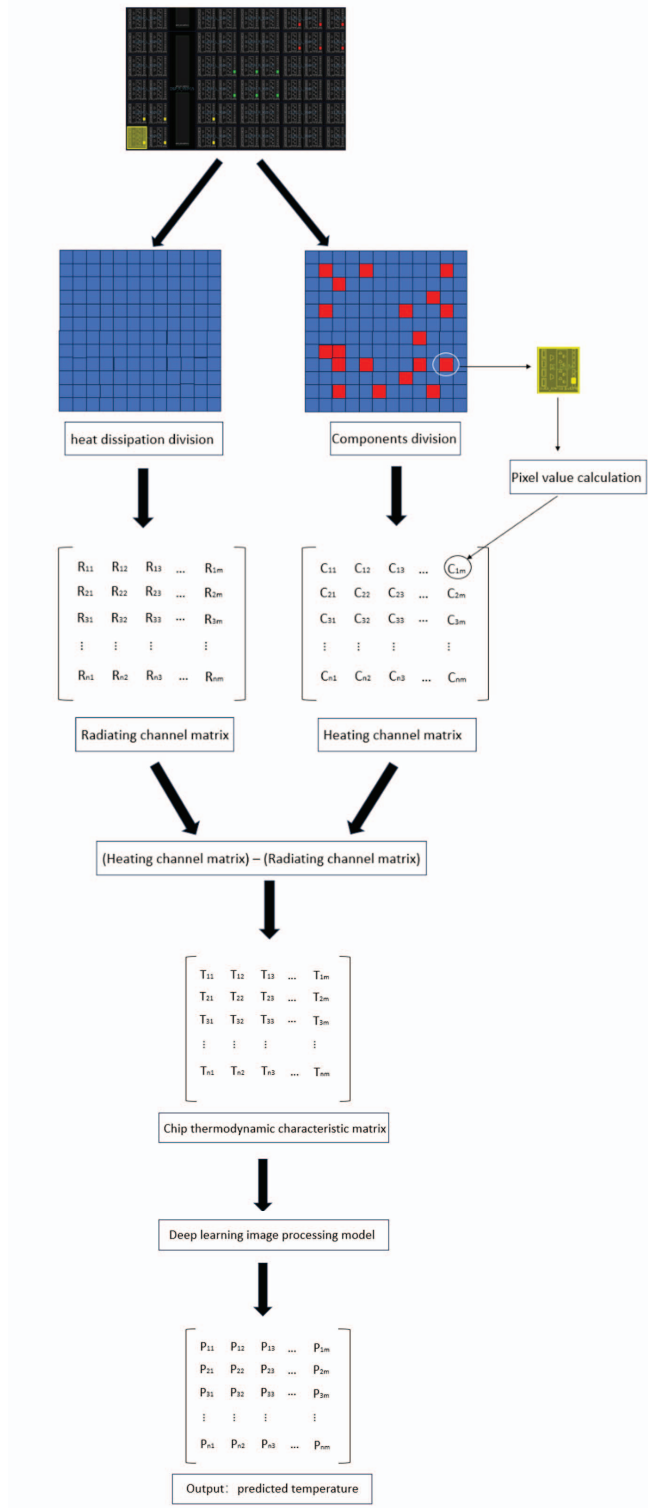


Fig. 1. The method of image-based modeling.

Due to the fact that the transistors of FPGA are placed in the macro, the entire chip layout is first segmented according to the size of the macro. In this segmentation, the macro is mainly analogized to a pixel in the image, and the corresponding size of the image matrix is obtained based on the number of macros inside the chip (directly selecting the segmentation component as the minimum pixel will cause the features of the pixel to be not clear enough).

Subsequently, based on a device on a macro, I divided it into two channels. The first represents the heating channel, and the matrix value of this channel is determined by the type and number of components inside the macro. The assignments for the macro with components are shown in Table I. For macro pixels without components, the feature value is set to 0.

TABLE I.
THE FACTORIZATION OF COMPONENT

Component	Corresponding factor
BUFGCE	2
LUT	3
FDRE	3
RAMB36E2	5
CARRY8	9
DSP48E2	17
IBUF	19
OBUF	19

The specific calculation formula is as follows (1), where the component and corresponding factor correspond to Table I above.

$$\sum(\text{component} * \text{corresponding factor}) = \text{macro_factor} \quad (1)$$

The second heat dissipation channel represents the heat dissipation capacity of this macro. As the data obtained does not have relevant information on heat dissipation fins, the matrix value of the heat dissipation channel in this article will be fixed to represent its own heat dissipation capacity.

Finally, the feature matrix of the heating channel is subtracted from the feature matrix of the cooling channel to obtain the final single channel input chip image matrix. The positive or negative internal feature values of this single channel matrix represent whether this macro is more in a heating or cooling state during chip operation.

III. EXPERIMENT

After obtaining the required experimental data, the next step is to attempt validation using mature deep learning algorithm models to ensure that the current functionality is met. In this process, I chose various networks that have shown good performance in image processing. After careful consideration, I have finally decided to use the following three excellent image processing neural networks.

Firstly, GoogleNet [12] is chosen. The most distinctive feature of GoogleNet, compared to other networks, is its use of the so-called Inception module. This is a multi-branch structure that can extract features simultaneously at different scales. Each Inception module contains different sizes of convolutional kernels and pooling operations to capture features at different scales. Since the modeling size of the image is less than 300 * 300, this article chooses the version of GoogleNet V1 as the representative of the GoogleNet network.

Secondly, ResNet [13], a deep learning network that allows stacking of deeper residual blocks is considered; In this article, Model ResNet50 is selected. ResNet introduces residual blocks that enable the network to learn residual mappings rather than complete mappings. This makes it easier for the network to learn identity mappings, facilitating the training of deep networks. The biggest difference between ResNet and GoogleNet is that one uses Inception blocks instead of convolution, while the other uses residual blocks, and the specific details of the two networks are shown in Fig. 2.

Lastly, the Feature Pyramid Network (FPN) [14], designed for extracting image features, is selected. FPN utilizes the structure of a feature pyramid and has achieved remarkable success in object detection tasks. Through its algorithm design, FPN can support the segmentation of very small image features. Fig. 3 below shows the details of FPN, which has an additional up sampling link compared to Resnet.

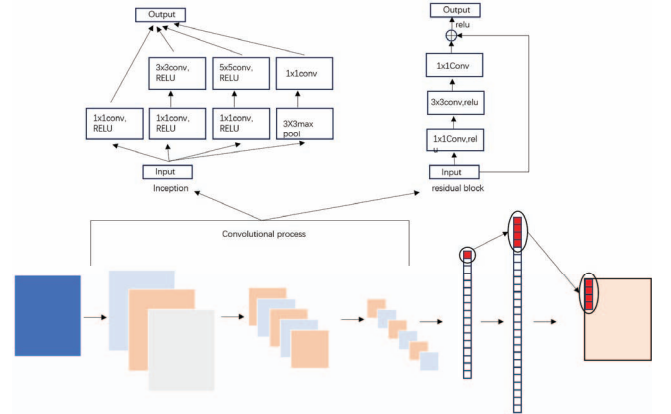


Fig. 2. Details of modified GoogleNet and ResNet.

According to the training data, this experiment can be divided into two parts. One is a data source with 5000 chip layouts with fewer transistors. Depending on the layout, the maximum operating temperature difference can reach 17 °C; The second batch of data includes chip layout data with over 50000 transistors and 1000 transistors. Due to the limitations of Hotspot, the minimum pixel count for this step of data includes 4 macro pixels, which also poses certain issues.

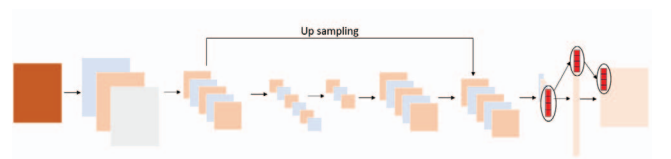


Fig. 3. Details of modified FPN.

IV. RESULT

A. Small scale chips

The results obtained from this experiment based on different networks are shown in TABLE II. The internal percentage refers to the ratio of nodes that accurately predict temperature after chip segmentation to the total number of points that need to be predicted. The specific visualization effect is shown in Fig. 4. In the figure, the temperature will be considered to zero in areas without transistors.

B. Large scale chips

This result is from a large-scale transistor chip. The layout data of this experiment is limited, and the node area division is too large. The specific results are shown in Table III. The internal percentage refers to the ratio of nodes that accurately predict temperature after chip segmentation to the total number of points that need to be predicted. The specific visualization effect is shown in Fig.5.

V. CONCLUSION

This article mainly explores the feasibility and new methods of applying deep learning in thermal analysis in the semiconductor field, and also provides an idea for exploring the feasibility of introducing new algorithms (such as AI drawing algorithms) in chip design. The final summary of this article is as follows:

- The combination of image-based modeling and deep learning can meet the needs of fast temperature prediction for chip layout. This method is accurate enough and provides a new direction for the application of deep learning in chip design. In addition to the new direction, this method also has a speed improvement compared to other methods. Hotspot requires more than 5 minutes to obtain data with this accuracy, while deep learning models only take 5 seconds.
- The method proposed in this article has poor performance in predicting the edge features of chip layout, which is related to directly predicting the demand of corresponding nodes. It can be attempted to use the method of predicting temperature field for thermal analysis.
- The size of chip node segmentation may also have an impact on the work of the model. Due to the limitation of hotspots, the node segmentation area of large-scale transistor chips is four times larger than that of small-scale transistor chips, resulting in excessive maximum error.

TABLE II.
SMALL SCALE CHIPS IN DIFFERENT NETWORKS

Deep Learning model	Temperature prediction error range and maximum error	
	Error within 1.5 °C	maximum
Google Net	100%	1.55°C
ResNet	100%	0.15°C
FPN	99.5%	3.6°C

TABLE III.
LARGE SCALE CHIPS IN DIFFERENT NETWORKS

Deep Learning model	Temperature prediction error range and maximum error	
	Error within 1.5 °C	maximum
Google Net	95.5%	85.1°C
ResNet	90.9%	197.8°C
FPN	16.4%	135.7°C

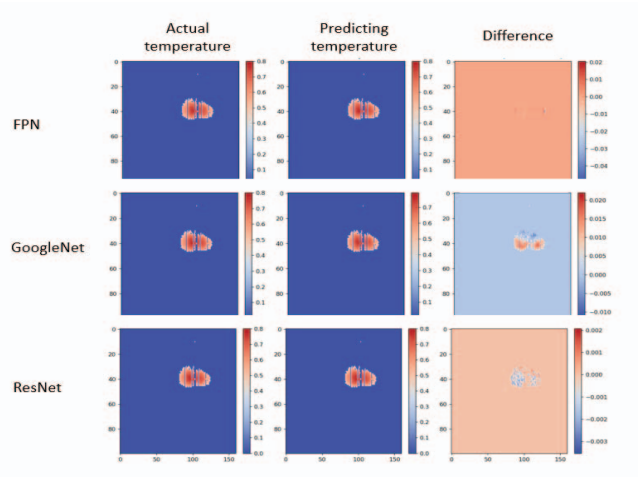


Fig. 4. Different network effects of small scale chips.

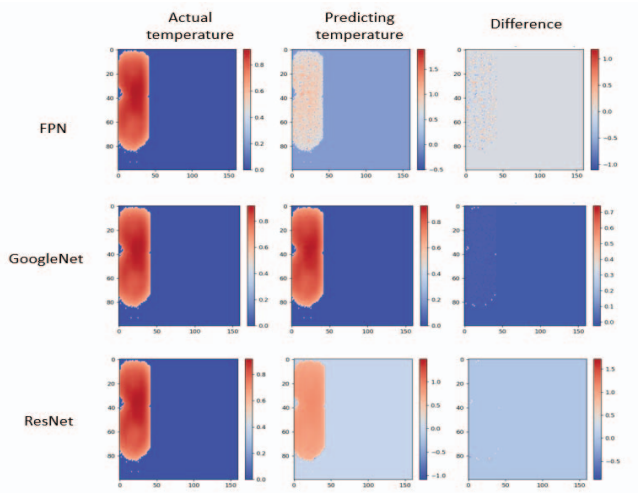


Fig. 5. Different network effects of large scale chips.

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